Graph-Masked Self Attention For Multi-Hop Reasoning

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Introduction

This paper explores combining Graph Neural Networks (GNNs) and Transformers by leveraging a masked attention mechanism to improve multi-hop reasoning on graph-structured data. By restricting attention via graph connectivity, the approach aims to utilize the Transformer's long-range capability while injecting graph inductive bias. However, further gains from scaling Transformers often require a rapid increase in computational cost. In contrast, GNNs struggle with long-range reasoning and deep message passing due to over-smoothing and over-squashing. This project targets this gap by overcoming GNN bottlenecks using Transformer-style attention while mitigating the Transformer's inefficiency through graph-like sparsity. The core idea is to use a graph-derived attention mask so the model focuses on graph-relevant connections and ignores distractors. If successful, this could enable more efficient and accurate reasoning in knowledge graphs, network analysis, and algorithmic problem-solving.

Background and Rationale

Modern Transformers face a fundamental challenge: dense self-attention has quadratic cost $O(n^2)$ in sequence length. Recent work attempts to scale attention to longer contexts by reducing complexity (Sun et al., 2025). FlashAttention improves practical speed and memory efficiency by avoiding explicit materialization of the attention matrix through IO-aware tiled computation, while still computing exact dense attention (Dao et al., 2022). However, the attention pattern remains dense and therefore quadratic, motivating sparse and structured alternatives.

In contrast, GNNs are designed for graph-structured data because they propagate messages along edges and naturally respect graph topology. While effective for local patterns,

challenges arise as the number of hops k increases: information from an exponentially growing neighborhood must pass through limited-capacity representations, leading to over-squashing (Alon & Yahav, 2020; Topping et al., 2021). Stacking many layers can also lead to over-smoothing, where node representations become increasingly similar and less distinguishable (Li et al., 2018; Oono & Suzuki, 2019). These bottlenecks make it difficult for message-passing GNNs to reliably capture long-range dependencies without loss or distortion.

One approach is attention-based GNNs, most notably Graph Attention Networks (GAT), which learn to weight neighbor messages rather than aggregating them uniformly (Veličković et al., 2017). However, GAT and message-passing GNNs remain fundamentally local, meaning deeper reasoning typically requires deeper networks, which can exacerbate these stability issues at higher hop counts.

There has also been growing interest in injecting graph inductive bias into Transformers via structural encodings, attention biases, or graph-derived sparsity. Graphormer adds graph-structure signals (e.g., shortest-path distance bias) to improve graph representation learning (Ying et al., 2021), while GraphiT uses diffusion-inspired mechanisms that decay interaction strength with graph distance (Mialon et al., 2021). More recent work explores explicitly sparse graph Transformers that reduce attention cost using structured sparsity and virtual nodes (Shirzad et al., 2023). Surveys organize these approaches into families such as attention bias, masking, token augmentation, and hybrid GNN–Transformer designs (Yuan et al., 2025).

Hypothesis

The methods will explore whether a Transformer with a graph-structured attention mask improves its ability to solve multi-hop relational reasoning tasks in the presence of irrelevant information. Concretely, a Transformer whose self-attention is restricted to follow the edges of a

known graph will achieve higher accuracy and filter out distractors more effectively than dense and window-mask attention. Moreover, graph-masked attention will use its attention capacity more efficiently with less connectivity, potentially leading to less computational cost.

Methods

A synthetic graph dataset is created to test this hypothesis. Each data sample consists of a directed graph with a designated start node v_0 and a query asking for the node reached by a k-hop path from v_0. The dataset is also constrained with unique k-hop path $v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_k$ that leads to correct destination of v_k. Additionally, a fixed number of distractors are added—extra directed edges between random nodes that are irrelevant to the true k-hop path. The difficulty is controlled by adjusting the number of distractor edges, which increases graph density and potential attention confusion. The model's responsibility is to read a list of all edges and the query "start node = X, hops = *k*" and output the correct target node. The input is a sequence of tokens representing the entire graph's adjacency list followed by the query, and output is the target node ID. This synthetic dataset is crucial as the ground-truth reasoning path is statically set and can generate unlimited examples with controlled complexity. For the purpose of this research dataset is set to 30 nodes, path length of $k = 3$ (three hops), and 20–40 distractor edges for a challenging scenario. As for the model architecture, a small Transformer classifier with token embeddings, L encoder layers ($L = 2$), and final linear head. The hidden state at the $[SEP]$ token (between the edge list and query) is used as a pooled representation and converted to score for each node ID. The model is trained with cross-entropy loss to predict the target node.

The key experimental factor is the attention mask. This is done through imposing three conditions while keeping everything else consistently throughout the experiment. The first one is a default dense attention (baseline), where any single token can be attended to any other token

freely. Dense (vanilla) attention is also noted as 100% connectivity. Next is local window attention with fixed window size of $w = 32$, meaning each token attends to at most 32 tokens around it in sequence order. Although the locality in the sequence is not a true depiction of graph locality, window attention is set as another baseline for an uninformed sparse method. Lastly, graph-structured attention is evaluated. The adjacency-based mask is applied such that tokens corresponding to an edge or node can only attend to tokens that share a node with them. In practice, a boolean attention mask is constructed $m \in 0,1^{T \times T}$ where T is the input sequence length. Here, $m_{ij} = 1$ blocks attention from token i to token j, while $m_{ij} = 0$ allows it. The allowed connectivity is reported as a fraction.

For training, AdamW optimizer is utilized with learning rate of 1e-3, with each epoch consisting of several thousand training examples. The training and validation split into 80/20 respective ratio of the entire dataset. Experiments were run in Google Colab (Python 3) using an NVIDIA T4 GPU.

Results and Preliminary Data

As the conditions were evaluated on a synthetic k-hop reachability task with an emphasis on $k = 3$, the performances were measured by validation accuracy. Specifically, the best validation accuracy achieved averaged over multiple random seeds for stability. Random chance is $1/N$ (~3.3% when N = 30).

# Distractor Edges	Dense Attention	Window Attention	Graph Attention
20	4.80%	5.47%	6.07%
40	7.27%	6.07%	9.73%
80	21.80%	5.60%	28.73%

Table 1. Mean across 5 random seeds for $k = 3$, reported as percentages.

Although absolute accuracies are low, consistent differences emerge across attention conditions. In Table 1, graph-masked attention achieves the highest validation accuracy at every distractor level, and its advantage grows as the graph becomes noisier. Table 2 shows that this improvement is achieved with substantially lower connectivity: graph-masked attention uses only ~7.2% → 2.5% of attention connections, compared to 100% for dense attention. Window attention reduces connectivity but remains consistently weak, suggesting sequence-local sparsity does not align with graph-local dependencies. In Figure 1, graph-masked attention also reaches its peak earlier than dense attention under the same setting.

# Distractors	Dense Allowed %	Window Allowed %	Graph Allowed %
20	100.0	47.16	7.20
40	100.0	27.48	4.21
80	100.0	14.91	2.49

Table 2. Mean allowed attention connectivity (%) across 5 random seeds for k = 3.

Analyzing both Table 1 and Table 2, they show a favorable trade-off between accuracy and efficiency. Dense attention uses full connectivity (100%) but does not achieve the best accuracy, while window attention reduces connectivity substantially (47% → 15%) yet remains consistently low in performance. In contrast, graph-masked attention attains the highest validation accuracy across all distractor levels while using only a small fraction of attention connections (~7.2% → 2.5%). This supports the core hypothesis that injecting graph structure helps the model ignore irrelevant edges and focus on valid connectivity for multi-hop reasoning.

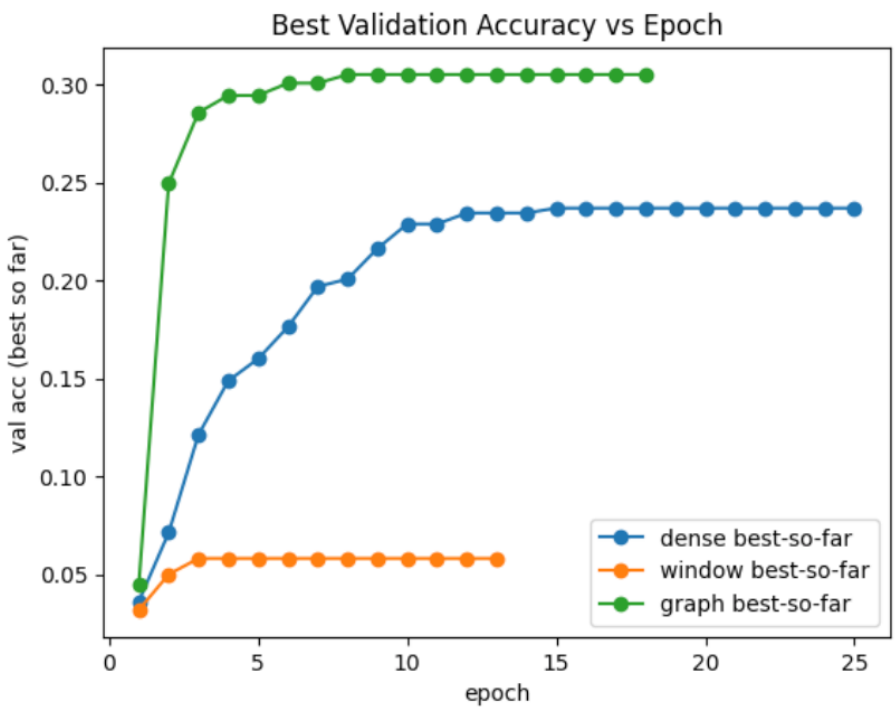

Figure 1. Best Validation Accuracy vs Epoch (k = 3, distractors = 40; *run_three()* example run).

In a single simulation (Figure 1), graph-masked attention improves rapidly in early epochs and reaches a higher peak accuracy (~0.305) than dense attention (~0.237), while window attention plateaus near ~0.058. This further supports that graph-consistent sparsity better captures multi-hop dependencies than sequence-local masking under the same difficulty setting.

Future Direction

Although this report emphasizes $k = 3$, the grid trends suggest the task becomes less stable as the number of hops increases. When k is set to 4 or 5, all attention conditions move closer to near-random chance across distractor settings, which is consistent with multi-hop reasoning becoming harder as k increases. A key next step is to test whether targeted architectural changes can preserve the benefits of graph masking when both distractors and hop count increase. Moreover, the plan is to explore deeper models with more Transformer layers, masks that allow limited multi-hop attention within a single layer. Testing distance-aware biases that

prioritize closer nodes while still permitting longer-range information flow is also noted for future direction. In parallel, scaling the dataset and adding GNN baselines (GCN/GAT) will help determine whether higher-*k* failures stem from limited capacity, overly strict masks, or task difficulty. It would also be interesting to investigate how Graph-of-Thought style inference, along with recent discovery on Recursive Language Models and Single-Source Shortest Paths could inform or extend graph-masked attention for multi-step reasoning (Besta et al., 2023; Duan et al., 2025; Zhang et al., 2025).

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